IN1006 Systems Architecture

Tutorial 02: Data Representation

Exercises

1. Complete Session 2 self-check online tutorial questions (Moodle)
2. What is the binary representation of 21, 74, 234
   0001 0101
   0100 1010
   1110 1010

3. What is the decimal equivalent of the unsigned binary number 1 0 0 1 0 1 0 1
   1 + 4 + 16 + 128 = 149

4. What is the largest unsigned number that can be stored in a 4-bit word? 8-bit word? 16-bit word? and a 32-bit word?
   2^4 – 1
   2^8 – 1
   2^16 – 1
   2^32 – 1

5. What is the one's complement of these binary numbers 10101100 and 01010100

01010011
10101011

6. Taking into account storage in 2's complement, what is the range of numbers (the largest and smaller) that could be stored in a 3 bit word? 4 bit word? and an 8 bit word?

011 = 3 is the largest number that can be stored in a 3 bit number
100 = -4 is the smallest number that can be stored in a 3 bit number

0111 = 7 is the largest number that can be stored in a 4 bit number
1000 = -8 is the smallest number that can be stored in a 4 bit number

2^n-1 -1 to –(2^n-1)

2^7 – 1 = 127
-(2^7) = -128

7. Negate (that is find the negative) of the binary unsigned number 01010011 using 2's complement arithmetic. The final number should be a binary in 2's complement.

1010 1100

1010 1101

8. Express each of the following two's complement representations into its equivalent decimal form.

00011	01111	11100	11010	10000
3	15	-4	-6	16

9. Convert the following decimal numbers into two's complement form using an 8 bit word size frame.
   a. 12 0000 1100
   b. -15 1111 0001
   c. -117 1000 1011
   d. -175 ERROR (Out of range)

10. Convert C6 hexadecimal representation of a two's complement binary number into decimal. Show both the two's complement binary representation and the decimal number.

C = 1100
6 = 0110

1100 0110
0011 1001
0011 1010
-58

11. Can the following operation on 2's Complement numbers produce overflow?
   a. 1??? + 1??? Yes
   b. 0??? – 0??? No

12. If we have an 8 bit word, what is the value of the Least Significant Bit when set? What about a 16 bit word? And a 32 bit word?

The value of LSB is always 1 when it is set regardless of the size of the word.

13. What is the binary representation of the following hexadecimal numbers: BC, 67, F0, 10, 3F. Give also their equivalent in decimal notation. Remember that one hexadecimal digit can represent 4 bits.

1011 1100 = 188
0110 0111 = 103
1111 0000 = 240
0001 0000 = 16
0011 1111 = 63

14. Express the following binary numbers in hexadecimal notation: 101010101010, 110010110111, 000011101011, 100100100100.

1100 1011 0111, 0000 1110 1011, 1001 0010 0100
C B 7, 0 E B, 9 2 4

15. What is the Octal representation of the following decimal numbers: 37, 21, 89

4 5

2 5

131

16. Convert these Octal numbers into hexadecimal notation: 17, 23, 56. . Remember that an octal digit can represent 3 bits.

001 111 = 0001 1111 = 1F

010 011 = 0001 0011 = 13

101 110 = 0010 1110 = 2E

17. Add this unsigned binary numbers: 01010101 + 00111111 and 10000111 + 00000011

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  01010101          10000111
+ 00111111        + 00000011
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  10010100          10001010
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18. Perform the following calculation, using two's complement addition to do the subtraction. The initial numbers should be considered as unsigned integers and the answer should be given in hexadecimal and decimal representation from the binary result.

$2EC_{16} - 3206_{8}$

2EC = 0010 1110 1100

3206 = 011 010 000 110

100 101 111 001

100 101 111 010

748 - 1670

```
  0010 1110 1100
+ 1001 0111 1010
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1 1000110 0110
```

− 922

19. Convert each of the following decimal representations to its equivalent two's complement form using 8 bit words; 6, -6, -17, 13, 0, -1

0000 0110

1111 1010

1110 1111

0000 0000

1111 1111

20. In the following problems, each bit word represents a value stored in two's complement notation. Find the answer to each problem in two's complement notation by performing the additions.

a.	b.	c.	d.	e.
0101	0011	0101	1110	1010
+0010	+0001	+ 1010	+0011	+1110
0111	0100	1111	0001	1000

21. Translate each of the following problem from decimal notation to into two's complement notation using bit words of length four then convert each problem to an equivalent addition problem(as a computer might do), and finally perform the additions. Check your answers by converting them back to decimal notation.

3	0011	4	0100	1	0001
-2	1110	-6	1010	-5	1011
1	0001	-2	1110	-4	1100

22. Perform the following calculation, using 2's complement addition to do the subtraction. Show all working. The initial numbers are in 2's complement and the answer should be given in binary 2's complement and in the decimal and the hex representation of the binary.

103A2B

0001 0000 0011 1010 0010 1011

010 110 011 010 000 001 101 000 → 0101 1001 1010 0000 0110 1000

1010 0110 0101 1111 1001 1000

  0001 0000 0011 1010 0010 1011
+ 1010 0110 0101 1111 1001 1000

1 011 0110 1001 1001 1100 0011

B 6 9 9 C 3

-4810301

$103A2B_{16}$ - 26320150_8